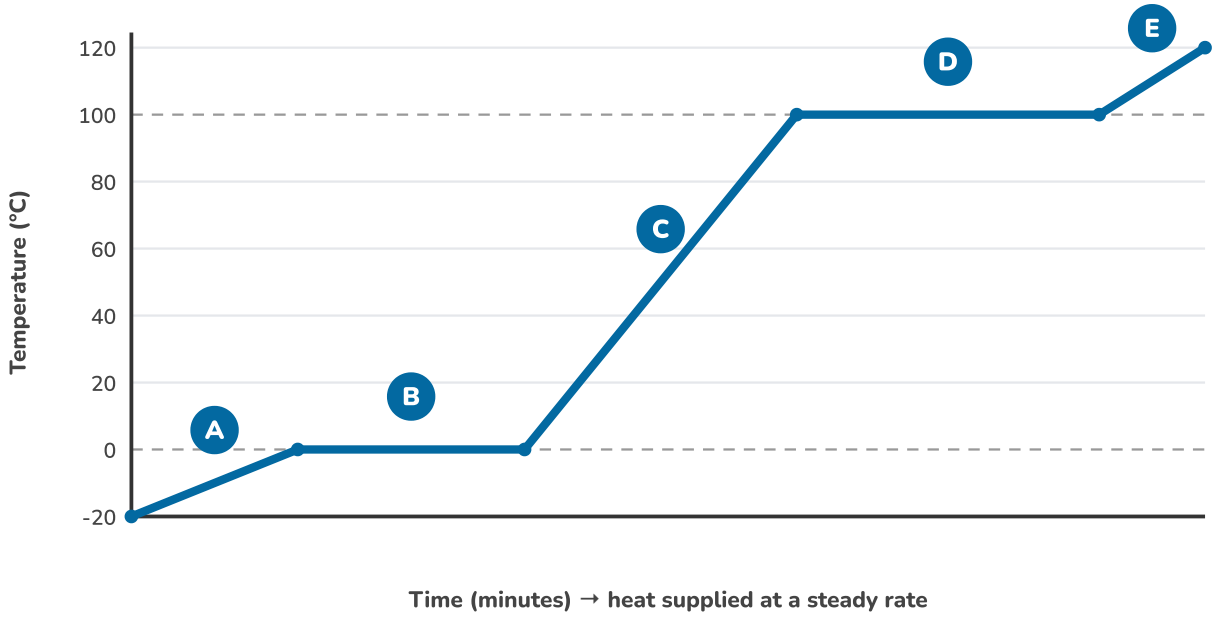
HEATING CURVE OF WATER

Name: ..... **Date:** .....

The graph shows a block of ice heated steadily until it has all become steam. Use it to answer the questions.



QUESTIONS

What state is the water in during section A?

What is happening during section B, and what is the temperature?

Heating continues during section B, but the temperature does not rise. Why?

In which section is the water a liquid only?

What boiling point does the graph show?

Which states are present during section D?

In which sections is the temperature rising?

The heater is switched off at the end of E and the steam is cooled. Describe the shape of the cooling graph.

HEATING CURVE OF WATER ANSWER KEY

Name: ..... **Date:** .....

QUESTIONS

1. Solid (ice), warming from −20 °C towards 0 °C. 2. The ice is melting; the temperature stays at 0 °C. 3. The energy is being used to break the forces holding the particles in fixed positions (solid to liquid), not to make the particles move faster. 4. C 5. 100 °C 6. Liquid and gas: the water is boiling. 7. A, C and E 8. The reverse: temperature falls, flat at 100 °C while the steam condenses, falls again, flat at 0 °C while the water freezes, then falls below 0 °C.